

Blood, Heart, and Circulation

Human Physiology Chapter 13

Biology Teaching Resources

December 24, 2025

Outline

13.1 Functions and Components of the Circulatory System

13.2 Composition of the Blood

13.3 Structure of the Heart

13.4 Cardiac Cycle

13.5 Electrical Activity of the Heart and the Electrocardiogram

13.6 Blood Vessels

13.7 Atherosclerosis and Cardiac Arrhythmias

13.8 Lymphatic System

Summary

Outline

13.1 Functions and Components of the Circulatory System

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13.6 Blood Vessels

13.7 Atherosclerosis and Cardiac Arrhythmias

13.8 Lymphatic System

Summary

13.1 Functions and Components of the Circulatory System

- The **circulatory system** (循环系统) has three main functions:
 - **Transportation** (运输): Carries O_2 , CO_2 , nutrients, wastes, and hormones
 - **Regulation** (调节): Maintains temperature and hormonal balance
 - **Protection** (保护): Prevents blood loss through clotting and provides immune defense
- Major components include the **heart** (心脏), **blood vessels** (血管), and **lymphatic system** (淋巴系统)

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13.2 Composition of the Blood

13.3 Structure of the Heart

13.4 Cardiac Cycle

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13.6 Blood Vessels

13.7 Atherosclerosis and Cardiac Arrhythmias

13.8 Lymphatic System

Summary

13.2 Composition of the Blood

- **Blood** (血液) consists of **plasma** (血浆) and **formed elements** (血细胞成分)
- Plasma is 92% water containing proteins, electrolytes, nutrients, gases, and wastes
- Formed elements include **red blood cells** (红细胞), **white blood cells** (白细胞), and **platelets** (血小板)



13.2 Composition of the Blood

- After centrifugation, blood separates into layers:
 - Top: Plasma (clear, straw-colored liquid, 55% of blood volume)
 - Middle: **Buffy coat** (白细胞层) - thin layer of white blood cells and platelets
 - Bottom: Packed red blood cells (45% of blood volume)
- The **hematocrit** (红细胞压积) measures the percentage of blood volume occupied by RBCs
- Normal hematocrit: Males 42-52%, Females 37-47%

13.2 Composition of the Blood

- **Erythrocytes** (红细胞) are the most abundant blood cells
- Contain **hemoglobin** (血红蛋白) for O_2 transport
- Biconcave disc shape increases surface area for gas exchange
- Lack nucleus and organelles, maximizing space for hemoglobin
- Lifespan: approximately 120 days



13.2 Composition of the Blood

- The biconcave disc shape provides:
 - Maximum surface area-to-volume ratio for efficient gas diffusion
 - Flexibility to squeeze through narrow capillaries
 - Optimal packing density in blood vessels
- Red blood cells are produced in **bone marrow** (骨髓) and destroyed in the spleen

13.2 Composition of the Blood

- **White blood cells** (白细胞) provide immune defense
- Five types with different functions:
 - **Neutrophils** (中性粒细胞): Phagocytize bacteria (50-70% of WBCs)
 - **Lymphocytes** (淋巴细胞): T cells and B cells for adaptive immunity (20-40%)
 - **Monocytes** (单核细胞): Become macrophages, phagocytize debris (2-8%)



13.2 Composition of the Blood

- **Eosinophils** (嗜酸性粒细胞):
Combat parasites and allergic reactions (1-4%)
- **Basophils** (嗜碱性粒细胞): Release histamine in inflammation (0.5-1%)

13.2 Composition of the Blood

- **Granular leukocytes** (粒细胞) contain visible cytoplasmic granules:
 - Neutrophils, eosinophils, basophils
- **Agranular leukocytes** (无粒细胞) lack visible granules:
 - Lymphocytes, monocytes
- **Platelets** (血小板) are cell fragments essential for blood clotting
(150,000-400,000 per μL)

13.2 Composition of the Blood

- **Hematopoiesis** (造血) is the formation of blood cells in red bone marrow
- All blood cells originate from **hematopoietic stem cells** (造血干细胞)
- **Erythropoiesis** (红细胞生成) is stimulated by **erythropoietin** (EPO, 促红细胞生成素)
- EPO is produced by kidneys in response to **hypoxia** (缺氧)



13.2 Composition of the Blood

- Stages of erythropoiesis:
 - Hematopoietic stem cell → Proerythroblast → Erythroblast → Normoblast → Reticulocyte → Mature erythrocyte
- Progressive changes include:
 - Decreasing cell size
 - Loss of nucleus
 - Increasing hemoglobin concentration
- **Reticulocytes** (网织红细胞) are immature RBCs still containing some RNA
- Iron, vitamin B₁₂, and **folic acid** (叶酸) are essential for normal RBC production

13.2 Composition of the Blood

- **ABO blood group system** (ABO 血型系统) determines blood type compatibility
- Based on presence or absence of A and B antigens on RBC surfaces
- Type A: A antigens, anti-B antibodies
- Type B: B antigens, anti-A antibodies
- Type AB: Both antigens, no antibodies (**universal recipient**, 万能受血者)
- Type O: No antigens, both antibodies (**universal donor**, 万能供血者)



13.2 Composition of the Blood

- **Agglutination** (凝集) occurs when antibodies bind to their corresponding antigens
- This causes red blood cells to clump together, blocking blood vessels
- Can be fatal if incompatible blood is transfused
- Type A blood mixed with anti-A antibodies → agglutination
- Type B blood mixed with anti-B antibodies → agglutination

13.2 Composition of the Blood

- Blood typing uses anti-A and anti-B antibodies to determine ABO blood type
- Add anti-A serum: Agglutination indicates A or AB blood
- Add anti-B serum: Agglutination indicates B or AB blood
- Both agglutinate: Type AB
- Neither agglutinates: Type O



13.2 Composition of the Blood

- **Cross-matching** (交叉配血) tests donor and recipient blood for compatibility
- **Rh blood group** (Rh 血型): Rh-positive (D antigen present) or Rh-negative (D antigen absent)
- **Hemolytic disease of the newborn** (新生儿溶血病) can occur when Rh-negative mother carries Rh-positive fetus
- Prevented by administering **RhoGAM** (Rh 免疫球蛋白) to Rh-negative mothers

13.2 Composition of the Blood

- **Hemostasis** (止血) prevents blood loss through three mechanisms:
 - **Vascular spasm** (血管痉挛): Immediate constriction of damaged vessel
 - **Platelet plug formation** (血小板栓形成): Platelets aggregate at injury site
 - **Blood coagulation** (血液凝固): Formation of fibrin clot
- Intact endothelium prevents platelet activation by secreting **nitric oxide**



13.2 Composition of the Blood

(NO, 一氧化氮) and **prostacyclin**
($\text{PG}_2^{\text{I}+}$, 前列环素)

13.2 Composition of the Blood

- When endothelium is damaged, **collagen** (胶原) is exposed
- Platelets adhere to collagen (**platelet adhesion**, 血小板黏附)
- Activated platelets release **ADP** (二磷酸腺苷) and **thromboxane A₂** (TXA₂)
- These substances cause more platelets to aggregate (**platelet aggregation**, 血小板聚集)
- Forms temporary **platelet plug** (血小板栓) that seals small breaks

13.2 Composition of the Blood

- Blood clotting forms a stable fibrin mesh that reinforces the platelet plug
- **Fibrin** (纤维蛋白) threads form a three-dimensional network
- Red blood cells become trapped in the fibrin mesh
- Platelets help contract the clot, pulling wound edges together



13.2 Composition of the Blood

- Colorized scanning electron micrograph shows:
 - Green: Fibrin threads forming the mesh structure
 - Red: Trapped erythrocytes
 - Purple: Platelets
- The fibrin network provides structural stability to the clot
- This prevents further blood loss while tissue repair occurs

13.2 Composition of the Blood

- Blood coagulation involves a cascade of enzymatic reactions
- Two pathways initiate clotting:
 - **Extrinsic pathway** (外源性途径):
Triggered by **tissue factor** (TF, 组织因子) from damaged tissue
 - **Intrinsic pathway** (内源性途径):
Triggered by contact with collagen
- Both converge in the **common pathway** (共同途径)



13.2 Composition of the Blood

- Key steps in the clotting cascade:
 - Both pathways activate factor X
 - Factor Xa converts **prothrombin** (凝血酶原) to **thrombin** (凝血酶)
 - Thrombin converts **fibrinogen** (纤维蛋白原) to **fibrin** (纤维蛋白)
- **Calcium ions** (Ca^{2+}) are required at multiple steps
- Many factors require **vitamin K** (维生素 K) for synthesis
- **Anticoagulants** (抗凝剂): Heparin, warfarin, aspirin

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13.8 Lymphatic System

Summary

13.3 Structure of the Heart

- The heart functions as two pumps working together:
 - Right side pumps blood through **pulmonary circulation** (肺循环)
 - Left side pumps blood through **systemic circulation** (体循环)
- Pulmonary circulation: Right ventricle → Lungs → Left atrium (oxygenation)
- Systemic circulation: Left ventricle → Body tissues → Right atrium (delivers O_2)



13.3 Structure of the Heart

- Key features of the dual circulation:
 - Pulmonary arteries carry deoxygenated blood (to lungs)
 - Pulmonary veins carry oxygenated blood (from lungs)
 - This is opposite to systemic circulation
- The heart has four chambers: Two atria (receiving chambers) and two ventricles (pumping chambers)
- **Interventricular septum** (室间隔) separates left and right ventricles

13.3 Structure of the Heart

- **Heart valves** (心脏瓣膜) ensure unidirectional blood flow
- **Atrioventricular (AV) valves** (房室瓣) between atria and ventricles:
 - **Tricuspid valve** (三尖瓣): Right side (3 cusps)
 - **Mitral valve** (僧帽瓣, bicuspid valve): Left side (2 cusps)
- **Semilunar valves** (半月瓣) at ventricular exits:
 - **Pulmonary valve** (肺动脉瓣): Right ventricle to pulmonary trunk



13.3 Structure of the Heart

- **Aortic valve** (主动脉瓣): Left ventricle to aorta

13.3 Structure of the Heart

- AV valves have **chordae tendineae** (腱索) attached to **papillary muscles** (乳头肌)
- These structures prevent valve prolapse during ventricular contraction
- Valves open and close passively in response to pressure differences
- No energy is required for valve operation

13.3 Structure of the Heart

- Semilunar valves have three crescent-shaped cusps
- Shown here in closed position, preventing backflow
- During diastole, blood in the arteries tries to flow back into ventricles
- The cusps fill with blood and seal the opening



13.3 Structure of the Heart

- When ventricles relax, arterial pressure exceeds ventricular pressure
- Blood flows backward, filling the valve cusps
- The cusps press together, closing the valve
- This prevents regurgitation of blood into the ventricles

13.3 Structure of the Heart

- **Heart sounds** (心音) are produced by valve closure
- First sound (S_1 , “lub”): AV valves close at start of **ventricular systole** (心室收缩)
- Second sound (S_2 , “dup”): Semilunar valves close at start of **ventricular diastole** (心室舒张)
- **Auscultation** (听诊) locations differ from anatomical valve positions due to sound transmission



13.3 Structure of the Heart

- **Heart murmurs** (心脏杂音) are abnormal sounds from turbulent blood flow
- Causes include:
 - **Valve stenosis** (瓣膜狭窄): Narrowed valve opening
 - **Valve insufficiency** (瓣膜关闭不全): Incomplete valve closure (regurgitation)
- Not all murmurs are pathological; some are functional or innocent murmurs

13.3 Structure of the Heart

- **Septal defects** (间隔缺损) are congenital heart defects
- Abnormal openings in the septum between chambers
- Cause left-to-right shunting (left side has higher pressure)
- (a) **Ventricular septal defect** (VSD, 室间隔缺损): Opening between ventricles
- (b) **Atrial septal defect** (ASD, 房间隔缺损): Opening between atria



13.3 Structure of the Heart

- Septal defects cause:
 - Mixing of oxygenated and deoxygenated blood
 - Volume overload of the pulmonary circulation
 - Increased workload on the heart
 - May lead to pulmonary hypertension if untreated
- Large defects require surgical repair

13.3 Structure of the Heart

- **Patent ductus arteriosus** (PDA, 动脉导管未闭) is failure of the **ductus arteriosus** (动脉导管) to close after birth
- In fetus: Ductus allows blood to bypass non-functional lungs
- Normally closes within days after birth
- If patent: Blood shunts from aorta to pulmonary artery



13.3 Structure of the Heart

- PDA causes:
 - Increased pulmonary blood flow
 - Volume overload of left heart
 - Continuous “machinery” murmur
- Treatment: Medication (indomethacin) in newborns or surgical ligation
- Early closure prevents complications

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Summary

13.4 Cardiac Cycle

- The **cardiac cycle** (心动周期) consists of all events in one heartbeat
- **Systole** (收缩期): Contraction phase
- **Diastole** (舒张期): Relaxation phase
- At 75 bpm, one cycle lasts 0.8 seconds



13.4 Cardiac Cycle

- Five phases of the cardiac cycle:
 1. **Atrial systole** (心房收缩): Atria contract, completing ventricular filling
 2. **Isovolumetric contraction** (等容收缩): All valves closed, pressure rises
 3. **Ventricular ejection** (心室射血): Semilunar valves open, blood ejected
 4. **Isovolumetric relaxation** (等容舒张): All valves closed, pressure drops
 5. **Ventricular filling** (心室充盈): AV valves open, passive filling
- At higher heart rates, diastole shortens more than systole

13.4 Cardiac Cycle

- Pressure changes drive valve opening and closing
- Events marked in the diagram:
 - (1) AV valves close → S_1 heart sound
 - (2) Semilunar valves open → ejection begins
 - (3) Semilunar valves close → S_2 heart sound
 - (4) AV valves open → filling begins



13.4 Cardiac Cycle

- Important volumes:
 - **End-diastolic volume** (EDV, 舒张末期容积): 135 mL (maximum filling)
 - **End-systolic volume** (ESV, 收缩末期容积): 65 mL (after ejection)
 - **Stroke volume** (SV, 每搏输出量): $EDV - ESV = 70 \text{ mL}$
 - **Ejection fraction** (射血分数): $SV/EDV = 52\%$ (normal $>50\%$)
- **Cardiac output** (CO, 心输出量) = $SV \times \text{Heart rate}$

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13.4 Cardiac Cycle

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13.6 Blood Vessels

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13.8 Lymphatic System

Summary

13.5 Electrical Activity of the Heart and the Electrocardiogram

- Cardiac muscle cells are **autorhythmic** (自律性):
Spontaneously depolarize
- **SA node** (窦房结) is the heart's natural **pacemaker** (起搏器)
- Located in right atrium near superior vena cava
- Depolarizes 70-80 times per minute at rest
- **Pacemaker potentials** (起搏电位) are spontaneous slow depolarizations



13.5 Electrical Activity of the Heart and the Electrocardiogram

- Pacemaker potentials caused by:
 - Decreased K^+ permeability (reduced outward current)
 - Increased Na^+ permeability (**funny current**, I_f)
 - Increased Ca^{2+} permeability
- When threshold (-40 mV) is reached, action potential is triggered
- SA node has fastest intrinsic rate, so it sets the heart rate

13.5 Electrical Activity of the Heart and the Electrocardiogram

- Cardiac action potentials differ from skeletal muscle
- Key feature: **Plateau phase** (平台期) maintained by slow Ca^{2+} influx
- Duration: 200-300 ms (vs. 1-2 ms in skeletal muscle)
- Long **refractory period** (不应期) prevents **tetanus** (强直收缩)
- Ensures adequate time for ventricular filling between contractions



13.5 Electrical Activity of the Heart and the Electrocardiogram

- Plateau phase mechanisms:
 - L-type calcium channels open slowly and remain open
 - Calcium influx balances potassium efflux
 - Membrane potential remains depolarized
- Repolarization occurs when calcium channels close and potassium efflux increases
- The long action potential ensures proper coordination of contraction

13.5 Electrical Activity of the Heart and the Electrocardiogram

- The **conduction system** (传导系统) coordinates heart contraction
- Pathway of impulse conduction:
 1. **SA node** (窦房结): Initiates impulse
 2. **Internodal pathways** (结间束): Conduct through atria
 3. **AV node** (房室结): Delays impulse 0.1 second
 4. **Bundle of His** (希氏束): Conducts through septum
 5. **Bundle branches** (束支): Right and left branches



13.5 Electrical Activity of the Heart and the Electrocardiogram

6. **Purkinje fibers** (浦肯野纤维):
Distribute throughout ventricles

13.5 Electrical Activity of the Heart and the Electrocardiogram

- AV node delay is crucial:
 - Allows atrial contraction to complete before ventricular contraction
 - Ensures ventricles are fully filled
 - Prevents premature ventricular contraction
- Purkinje fibers conduct rapidly (4 m/s)
- This ensures near-simultaneous ventricular contraction for efficient pumping

13.5 Electrical Activity of the Heart and the Electrocardiogram

- Correlation between action potential and contraction
- Long plateau phase corresponds to contraction phase
- Long refractory period prevents premature re-excitation
- No summation or tetanus possible in cardiac muscle
- This is essential for proper pump function



13.5 Electrical Activity of the Heart and the Electrocardiogram

- The action potential must end before relaxation can occur
- Mechanical events (contraction/relaxation) lag behind electrical events (depolarization/repolarization)
- This ensures coordinated, sequential pumping action
- Each contraction must complete before the next can begin

13.5 Electrical Activity of the Heart and the Electrocardiogram

- The **electrocardiogram** (ECG, 心电图) records the heart's electrical activity
- Three major waves:
 - **P wave** (P 波): Atrial depolarization
 - **QRS complex** (QRS 波群):
Ventricular depolarization
 - **T wave** (T 波): Ventricular repolarization
- Atrial repolarization is hidden within the QRS complex



13.5 Electrical Activity of the Heart and the Electrocardiogram

- Important intervals:
 - **P-R interval** (P-R 间期): 0.12-0.20 s (conduction time through AV node)
 - **Q-T interval** (Q-T 间期): Total ventricular depolarization and repolarization time
 - **S-T segment** (S-T 段): Should be at baseline (isoelectric)
- Electrical events precede and trigger mechanical events
- P wave → atrial contraction; QRS → ventricular contraction

13.5 Electrical Activity of the Heart and the Electrocardiogram

- The sequence of impulse conduction generates ECG waveforms
- Depolarization spreads from endocardium to epicardium
- Different time points show different stages of ventricular activation
- The direction and magnitude of electrical activity determine wave morphology



13.5 Electrical Activity of the Heart and the Electrocardiogram

- Conduction sequence:
 - SA node fires → atrial depolarization → P wave
 - AV node delay → P-R interval
 - Bundle branches and Purkinje fibers → rapid ventricular depolarization → QRS
 - Ventricular repolarization → T wave
- Multiple leads provide different electrical perspectives

13.5 Electrical Activity of the Heart and the Electrocardiogram

- **ECG leads** (导联) provide different views of the heart
- **Bipolar limb leads** (双极肢体导联): I, II, III (Einthoven's triangle)
- **Unipolar chest leads** (单极胸导联): V_1 - V_6 (precordial leads)
- 12-lead ECG is standard for comprehensive evaluation



13.5 Electrical Activity of the Heart and the Electrocardiogram

- Different leads help localize:
 - Ischemia or infarction location
 - Conduction abnormalities
 - Chamber enlargement
- Chest leads positioned at specific intercostal spaces
- Each lead has characteristic normal waveform patterns

13.5 Electrical Activity of the Heart and the Electrocardiogram

- Correlation between electrical and mechanical events
- QRS complex triggers ventricular contraction → pressure rises (systole)
- T wave represents repolarization → relaxation follows (diastole)
- S-T segment corresponds to plateau phase of action potential



13.5 Electrical Activity of the Heart and the Electrocardiogram

- Timing relationships:
 - Electrical depolarization precedes contraction by 0.02 s
 - Electrical repolarization precedes relaxation
 - This lag is called **electromechanical coupling** (电-机械耦联)
- Understanding this relationship is crucial for interpreting ECG in clinical context

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13.8 Lymphatic System

Summary

13.6 Blood Vessels

- Blood vessel structure varies by function
- All vessels have three layers (tunicas):
 - **Tunica intima** (内膜): Endothelium lining
 - **Tunica media** (中膜): Smooth muscle and elastic tissue
 - **Tunica externa** (外膜): Connective tissue
- Arteries have thicker walls than veins to withstand higher pressure



13.6 Blood Vessels

- **Arteries** (动脉): Thick walls, narrow lumens, more elastic tissue
 - Carry blood away from heart under high pressure
- **Veins** (静脉): Thin walls, large lumens, less elastic tissue
 - Return blood to heart under low pressure
 - Many have valves to prevent backflow
- Wall thickness proportional to pressure

13.6 Blood Vessels

- **Microcirculation** (微循环) is the network of small vessels
- **Arterioles** (小动脉) regulate blood flow to capillary beds
- **Precapillary sphincters** (毛细血管前括约肌) control flow through individual capillaries
- **Metarterioles** (后微动脉) provide **arteriovenous anastomoses** (动静脉吻合)



13.6 Blood Vessels

- Blood flow regulation based on tissue metabolic needs:
 - Active tissues: Sphincters relax, capillaries open
 - Resting tissues: Sphincters contract, blood bypasses via metarterioles
- **Autoregulation** (自身调节): Local control of blood flow
- **Vasodilation** (血管舒张) increases flow; **vasoconstriction** (血管收缩) decreases flow

13.6 Blood Vessels

- Capillaries are exchange vessels
- Wall structure: Single layer of endothelial cells
- Diameter: 5-10 μm (barely wide enough for RBCs)
- Thin wall ($<1 \mu\text{m}$) allows efficient diffusion
- Total surface area: $\sim 600 \text{ m}^2$



13.6 Blood Vessels

- Exchange mechanisms at capillaries:
 - **Diffusion** (扩散): Primary mechanism for gases and small molecules
 - **Filtration** (滤过): Pressure-driven fluid movement out
 - **Reabsorption** (重吸收): Osmotic-driven fluid movement in
 - **Transcytosis** (胞吞胞吐): Vesicular transport for large molecules
- **Pinocytotic vesicles** (胞饮小泡) transport materials across endothelial cells

13.6 Blood Vessels

- Veins return blood to the heart against gravity (in lower body)
- **Venous valves** (静脉瓣) prevent backflow
- **Skeletal muscle pump** (骨骼肌泵): Muscle contractions compress veins
- Valves ensure blood flows only toward the heart
- One-way valve system prevents backward flow



13.6 Blood Vessels

- Mechanisms promoting venous return:
 - Skeletal muscle pump (especially in legs)
 - **Respiratory pump** (呼吸泵): Breathing assists venous flow
 - Sympathetic **venoconstriction** (静脉收缩) reduces venous capacity
 - Venous valves prevent backflow
- **Varicose veins** (静脉曲张): Enlarged veins with incompetent valves

13.6 Blood Vessels

- William Harvey's historic demonstration of venous valves (1628)
- By blocking venous drainage with tourniquet, he showed:
 - Blood in veins flows only toward the heart
 - Valves prevent backward flow
- This experiment helped establish the concept of circulation



13.6 Blood Vessels

- Harvey's observations:
 - Veins bulge distal to tourniquet (blood accumulates)
 - Pressing on vein empties it, but blood cannot flow backward
 - Valves act as one-way gates
- Revolutionary concept for understanding cardiovascular physiology

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13.7 Atherosclerosis and Cardiac Arrhythmias

13.8 Lymphatic System

Summary

13.7 Atherosclerosis and Cardiac Arrhythmias

- **Atherosclerosis** (动脉粥样硬化) is buildup of fatty deposits in arterial walls
- Plaque formation:
 - Endothelial injury → LDL accumulation
 - Macrophages engulf oxidized LDL → **foam cells** (泡沫细胞)
 - Smooth muscle proliferation
 - Lipid core with fibrous cap



13.7 Atherosclerosis and Cardiac Arrhythmias

- Consequences of atherosclerosis:
 - Narrowed lumen → reduced blood flow
 - Plaque rupture → **thrombosis** (血栓形成)
 - **Myocardial ischemia** (心肌缺血) or infarction
 - **Angina pectoris** (心绞痛): Chest pain from ischemia
- Major cause of coronary artery disease and heart attacks

13.7 Atherosclerosis and Cardiac Arrhythmias

- **Lipoproteins** (脂蛋白) transport cholesterol and triglycerides
- Structure: Lipid core surrounded by protein-phospholipid shell
- **Apolipoproteins** (载脂蛋白) serve as ligands and enzyme activators
- Core contains nonpolar lipids (triglycerides, cholesterol esters)



13.7 Atherosclerosis and Cardiac Arrhythmias

- Types of lipoproteins:
 - **LDL** (低密度脂蛋白): Delivers cholesterol to tissues (“bad cholesterol”)
 - **HDL** (高密度脂蛋白): Removes cholesterol from tissues (“good cholesterol”)
 - **VLDL** (极低密度脂蛋白): Transports triglycerides from liver
- High LDL and low HDL increase atherosclerosis risk

13.7 Atherosclerosis and Cardiac Arrhythmias

- **Myocardial ischemia** (心肌缺血) can be detected by ECG changes
- **S-T segment depression** (S-T 段压低): Subendocardial ischemia
- **S-T segment elevation** (S-T 段抬高): Transmural ischemia/infarction
- ECG changes help localize affected coronary artery



13.7 Atherosclerosis and Cardiac Arrhythmias

- Clinical significance:
 - S-T depression during exercise stress test indicates ischemia
 - S-T elevation in acute MI requires immediate intervention
 - Different leads show changes in different heart regions
- Treatment: Restore blood flow (angioplasty, stents, bypass surgery)

13.7 Atherosclerosis and Cardiac Arrhythmias

- **Arrhythmias** (心律失常) are abnormal heart rhythms
- (a) **Sinus rhythm** (窦性心律): Normal or abnormal rates
 - **Bradycardia** (心动过缓): < 60 bpm
 - **Tachycardia** (心动过速): > 100 bpm
- (b) **Atrial flutter** (心房扑动): Rapid atrial contractions (250-350 bpm)
- (c) **Ventricular tachycardia** (室性心动过速): Life-threatening rapid ventricular rhythm



13.7 Atherosclerosis and Cardiac Arrhythmias

- Other important arrhythmias:
 - **Atrial fibrillation** (心房颤动): Irregular atrial activity, risk of stroke
 - **Ventricular fibrillation** (心室颤动): Chaotic activity, no effective pumping, requires immediate defibrillation
 - **Premature contractions** (过早搏动): Extra beats from ectopic foci
- Treatment: Antiarrhythmic drugs, cardioversion, defibrillation, pacemakers, ablation

13.7 Atherosclerosis and Cardiac Arrhythmias

- **AV node block** (房室传导阻滞)
impairs conduction from atria to ventricles
- **First-degree block** (一度): Prolonged P-R interval (>0.20 s), all impulses conducted
- **Second-degree block** (二度): Some P waves not followed by QRS, dropped beats
- **Third-degree block** (三度, complete block): No relationship between P and QRS, independent rhythms



13.7 Atherosclerosis and Cardiac Arrhythmias

- Clinical implications:
 - First-degree: Usually benign, monitoring required
 - Second-degree: May progress to complete block
 - Third-degree: Requires pacemaker implantation
- AV block can result from MI, medication, or degenerative disease

Outline

13.1 Functions and Components of the Circulatory System

13.2 Composition of the Blood

13.3 Structure of the Heart

13.4 Cardiac Cycle

13.5 Electrical Activity of the Heart and the Electrocardiogram

13.6 Blood Vessels

13.7 Atherosclerosis and Cardiac Arrhythmias

13.8 Lymphatic System

Summary

13.8 Lymphatic System

- The **lymphatic system** (淋巴系统) maintains fluid balance and provides immune defense
- Functions:
 - Returns excess interstitial fluid to blood circulation
 - Transports dietary lipids from intestines
 - Provides immunological surveillance and defense



13.8 Lymphatic System

- **Lymphatic capillaries** (淋巴毛细管) are blind-ended vessels
- Highly permeable structure allows entry of:
 - Excess interstitial fluid
 - Proteins that leaked from blood capillaries
 - Pathogens and foreign particles
- Overlapping endothelial cells act as one-way valves

13.8 Lymphatic System

- The lymphatic system complements the cardiovascular system
- Pathway: Tissues → Lymphatic capillaries → Lymphatic vessels → Lymph nodes → Lymphatic ducts → Blood circulation
- **Thoracic duct** (胸导管): Drains most of body
- **Right lymphatic duct** (右淋巴导管): Drains right upper body



13.8 Lymphatic System

- Lymph returns to blood at the junction of subclavian and internal jugular veins
- Mechanisms of lymph flow:
 - **Skeletal muscle pump** (骨骼肌泵)
 - **Respiratory pump** (呼吸泵)
 - Smooth muscle contractions in lymphatic vessels
 - One-way valves prevent backflow
- No central pump like the heart

13.8 Lymphatic System

- **Lymph nodes** (淋巴结) are small, bean-shaped structures
- Functions:
 - Filter lymph, removing pathogens and debris
 - House lymphocytes and macrophages
 - Sites where immune responses are initiated
- Major clusters: Cervical, axillary, inguinal, abdominal regions



13.8 Lymphatic System

- Clinical significance:
 - **Lymphadenopathy** (淋巴结病): Enlarged lymph nodes during infection
 - Lymph nodes become swollen and tender as immune cells proliferate
 - **Edema** (水肿): Swelling from excess interstitial fluid
 - **Lymphedema** (淋巴水肿): Swelling from lymphatic obstruction
- Palpation of lymph nodes is important in physical examination

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Summary

Summary

- The circulatory system consists of the heart, blood vessels, and lymphatic system
- Blood serves transportation, regulation, and protection functions
- The heart pumps blood through pulmonary and systemic circulations
- Electrical activity of the heart coordinates mechanical pumping
- Blood vessels are specialized for their functions: arteries (transport under pressure), capillaries (exchange), veins (return)
- The lymphatic system maintains fluid balance and provides immune defense
- Understanding cardiovascular physiology is essential for clinical medicine